SOLANKI OM NARENDRA

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Ambitious AI/ML Undergraduate at Jain University specializing in the technical architecture of intelligent systems. Expert in bridging Machine Learning research with scalable Full-Stack engineering, with a focused arsenal in predictive modeling, dimensionality reduction, and advanced system optimization.

TECHNICAL ARSENAL & CORE COMPETENCIES

- **AI & Machine Learning:** Neural Architectures, NLP, Computer Vision, OpenCV, MediaPipe, TensorFlow, Scikit-learn (LDA, Autoencoders).
- **Data & Systems:** Data Warehousing, Virtualization, Database Management, Algorithms on Graphs, SQL, Linux, Docker.
- **Programming:** Python (Basics), Java , HTML , Git.
- Core Computer Science: Data Structures & Algorithms, OOP, Dynamic Programing , Data Analysis

DEEP TECH DEVELOPMENT

1. **CTI Event Classification over Sysmon Logs(Jun-2025)**
   - This project transforms millions of raw **Microsoft Sysmon logs** into high-priority security intelligence by treating system actions as a language. We began by extracting critical telemetry—such as **EventIDs, CommandLines, and ProcessGuids**—which provide a granular look at system behavior far beyond what standard antivirus tools can see. These text-based logs were then converted into numerical vectors using **NLP techniques**, allowing us to treat a sequence of system actions like a "sentence" that tells the story of an attacker's movement through a network.
   - To find hidden threats, we applied **Latent Dirichlet Allocation (LDA)** to discover "latent topics," which in this context represent complex cyber-attack strategies. By grouping these behaviors and mapping them to the **MITRE ATT&CK Framework**, the system can automatically identify specific tactics like *Privilege Escalation* or *Lateral Movement*. Finally, we integrated **Neural Autoencoders** to learn the mathematical "shape" of normal system activity; any log sequence that deviates from this baseline triggers a high reconstruction error, flagging it as a potential breach for immediate human interventio
2. **OpenCV & Computer Vision Specialized Projects**
   - **(A.) Dual-Threshold Face Recognition System(Sep 2025)**
     - This project implements a robust biometric security layer by processing high-resolution video streams to ensure precise identity verification. We utilized **OpenCV** to perform real-time face detection and landmark extraction, transforming complex visual features into 128-dimensional **Embedding Vectors**—essentially creating a unique numerical "fingerprint" for every authorized user in the database.
     - To solve the common security vulnerabilities of standard facial recognition, we architected a **Proprietary Dual-Threshold Mechanism**. This logic operates in two mathematical layers: an initial threshold for rapid broad identification and a secondary, high-stringency threshold for final verification. This specific design is engineered to drastically minimize the **False Acceptance Rate (FAR)**, effectively preventing "look-alikes" or high-quality photographic spoofs from bypassing the system. The result is a high-performance access control solution that balances computational speed with the extreme accuracy required for high-risk corporate or governmental environments.
   - **(B.) VisionEdge: Real-Time Multithreaded Emotion Analysis(Sep 2025)**
     - This project optimizes the deployment of heavy deep-learning models on standard hardware by engineering a high-performance **Edge AI** pipeline. We processed a continuous **60 FPS webcam feed**, utilizing the **DeepFace framework** to detect facial regions and quantify the intensities of seven core emotions. To maintain real-time efficiency without a dedicated GPU, we implemented **AI Scaling** to reduce the frame size by 50% and **Temporal Sampling**, which limits AI analysis to every 10th frame, significantly lowering the computational overhead per second.
     - The core innovation lies in the **Multithreaded Inference Engine**, which decouples the "Seeing" (video rendering) from the "Thinking" (AI processing) into separate CPU threads to eliminate display lag. Because the AI performs its analysis on downscaled frames, we developed custom **Coordinate Mapping logic** to mathematically "upscale" the detection boxes back onto the high-resolution live feed. This architectural approach ensures a "Status: Smooth" user experience, proving that complex vision systems can be both responsive and accurate on consumer-grade laptops.
3. Health-Predictive Analytics (ODE + ML) (jan 2026)
   - Developed a "Deep Tech" diagnostic engine by solving **Ordinary Differential Equations (ODEs)** via SciPy's odeint to model continuous heart rate dynamics (dHR/dt) instead of static data points. This mathematical approach simulates biological **Recovery Curves**, which are then processed through a **Logistic Regression** pipeline—integrated with BMI and Sleep metrics—to accurately forecast risks for Hypertension and Sleep Apnea based on physiological shifts.
4. Non-Contact Nurse-Call System (Feb 2026)
   - I spearheaded the Machine Learning and Data Intelligence layer, where I engineered the logic to transform raw gesture telemetry from the PAJ7620 sensor into prioritized medical alerts. My work focused on developing a robust classification framework to distinguish between accidental movements and intentional patient signals, ensuring high-fidelity input for the system's contactless interface. By optimizing the data flow through an MQTT broker to a Raspberry Pi server, I successfully integrated these live sensor streams into a predictive framework that monitored request patterns while maintaining the sub-100ms latency required for critical emergency communications. This "Deep Tech" healthcare solution effectively bridges IoT hardware with intelligent data processing to eliminate physical contact surfaces and reduce infection rates in hospital environments.
5. **Mini Projects (Jan 2025)**
   - **Email Spam Detection:** Utilized **TF-IDF Vectorization** and **Multinomial Naive Bayes** to classify raw text data into spam or legitimate categories based on word frequency.
   - **House Price Prediction:** Applied **Linear Regression** and **One-Hot Encoding** to historical real estate datasets to forecast property values by minimizing residual errors.
   - **Stock Price Analytics:** Engineered a real-time pipeline using the **Alpha Vantage API** and **mplfinance** to visualize market trends and predict future price movements.

EDUCATION AND LEADERSHIP & EXPERIENCE

- **B.Tech in Artificial Intelligence & Machine Learning: Pursuing a specialized degree at Jain University (CSE Department) with a core focus on NLP, Computer Vision, and IoT integration.**
- **Team Lead (ReGenX):** Spearheaded the Machine Learning and Data Intelligence layer for a national-level hackathon project, managing real-time telemetry and sub-100ms emergency alert prioritization.
- **Tech Team Member (The Turing Club):** Collaborated on the development and deployment of advanced AI solutions and high-performance software architectures within a select technical group.

CORE TECHNICAL CERTIFICATIONS

- **TensorFlow Developer Professional Certificate, DeepLearning.AI TensorFlow Developer, Microsoft Certified: Azure AI Engineer Associate, AWS Certified Machine Learning – Specialty, NVIDIA Deep Learning Institute (DLI) Certifications, CompTIA Security+, and Google Cloud Professional Machine Learning Engineer.Networking Foundations: Networking Basics , Introduction to NLP and LLMs: Principles and Practical Applications , and Machine Learning Statistical Foundations Professional Certificate.**